

QNu Labs is a great example of an Indian quantum startup. In fact, it is India's only quantum-tech cybersecurity company as of now. It is trying to reimagine cybersecurity in a futuristic way, build it here in India, and then take it to the whole world.

Another example would be BosonQ, a Bhilai based startup founded in 2021. The company creates cutting-edge software for quantum computing, such as multidisciplinary optimisation, computational aeroacoustics, computational fluid dynamics, computational structure dynamics, and computational heat transfer.

On the research front, a collaborative declaration to create a virtual network centre for quantum computing was signed by the governments of India and Finland on April 18, 2022. Even IBM is planning to create a national quantum plan in India. Institutions like TCS, DRDO, TIFR and many educational

Metric	Significance
Number of Qubits	Problem size capabilities
FTQEC Qubit Overhead	Number of additional qubits required to implement FTQEC
FTQEC Gate Overhead	Number of additional gates required to implement FTQEC
Clock Cycles Per Gate	Basis for determining quantum computation time
FTQEC Latency	Additional computation time needed to incorporate FTQEC
Number of Gates	Number of gates used to complete circuit
Sample Rate	The life-cycle frequency to obtain a result and reset a circuit to obtain an additional result
Entanglement Latency	Time needed to initially entangle a set of qubits
Entanglement Rate	Time needed to refresh or reestablish entanglement between a set qubits
Intra-QPU Connectivity	The number of qubits a single qubit can entangle with directly
Inter-QPU Connectivity	The number of qubits a single qubit can entangle with indirectly or at a remote site
Teleportation Rate	The rate at which quantum data can be exchanged
ISA Complexity	The diversity of gates that can being executed

Metrics of a QPU (Credit: Keith A. Britt and Travis S. Humble, ACM Journal on Emerging Technologies in Computing Systems, Vol. 1, No. 1, Article 1)

institutions are also working towards the same broad goal—developing a quantum ecosystem in India.

In a relatively short period of time, India certainly seems to be advancing its capabilities in quantum technology. And there's a lot more to come. **END** 🐧

By: Aaryaa Padhyegurjar

The author is an Industry 4.0 enthusiast at EFY with a keen interest in innovation and research.

The article was originally published in the November 2022 issue of Electronics For You.

www.opensourceforu.com

Open Source Is **Hot** In The IT World

Technologies shown: Hadoop, Apache, PostgreSQL, Android, MySQL, OSS, PEARL, OpenStack, Joomla, Drupal, PHP.

EFY GROUP

THE COMPLETE MAGAZINE ON OPEN SOURCE



- >> To find dealers to buy copy from news-stand, visit: www.ezine.lfyimag.com
- >> To subscribe, visit: pay.efyindia.com
- >> To buy an ezine edition, visit: www.magzter.com choose Open Source For You